



ENGINEERING MATHEMATICS - II

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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 1

Sub Topic : Introduction to Fourier Series

Department of Science and Humanities

1	UNIT-V <u>FOURIER SERIES</u>	Introduction to Fourier Series, Dirichlet's conditions, Euler's formula
2		Fourier series of even and odd functions
3 - 4		Problems on Fourier series
5		Half range Fourier series and problems
6		Parseval's identity
7 - 8		Practical harmonic analysis
9 - 10		Complex form of Fourier series

What is Fourier series?

Why do we need Fourier series?

What are the different types of Fourier Series?

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Introduction to Fourier Series

Fourier series is named after the **French Physicist Jean Baptise Fourier (1768 -1830)** who was the first to suggest that periodic signals could be represented by a sum of sinusoids.



What is Fourier Series?

In mathematics, **Fourier series decomposes any periodic function or periodic signal** into the sum of a possibly infinite set of simple oscillating functions namely sines and cosines (or complex exponentials)

Fourier series expansion is used to represent **a periodic signal** in terms of complex exponentials, or equivalently in terms of sine and cosine waveforms.

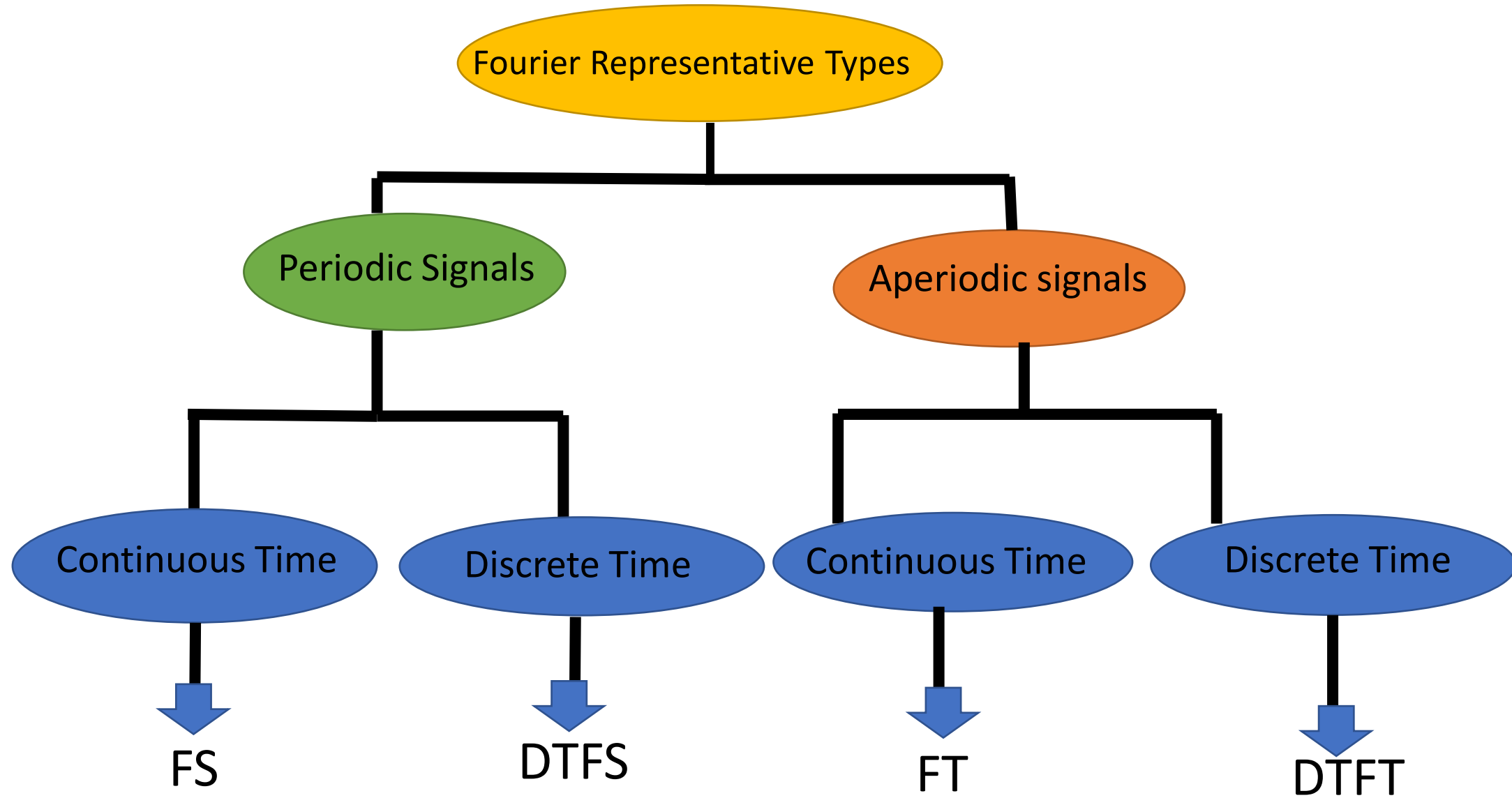
Why do we need Fourier series?

The primary reason that we use Fourier series is that we can better **analyze a signal** in another domain rather in the original domain.

What are the different types of Fourier Series?

The two types of Fourier series are - Trigonometric and exponential. The exponential is more convenient for Fourier series calculations.

Types of Fourier Representation



Applications of Fourier Series in Mathematics



- Fourier series are widely used to solve various ordinary differential equations (ODEs) and partial differential equations (PDEs).
- The Fourier series of functions in the differential equation often gives some prediction about the behaviour of the solution of differential equation. They are useful to find out the dynamics of the solution.

Applications of Fourier Series in Science and Engineering



- Fourier series has its applications in the field of digital signal processing, wave forms of electrical field, vibrations and conduction of heat etc.
- Also Fourier series finds its applications in electrical engineering, vibration analysis, acoustics, optics, signal processing, image processing, quantum mechanics, econometrics, thin-walled shell theory etc.

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Applications of Fourier Series in Science and Engineering

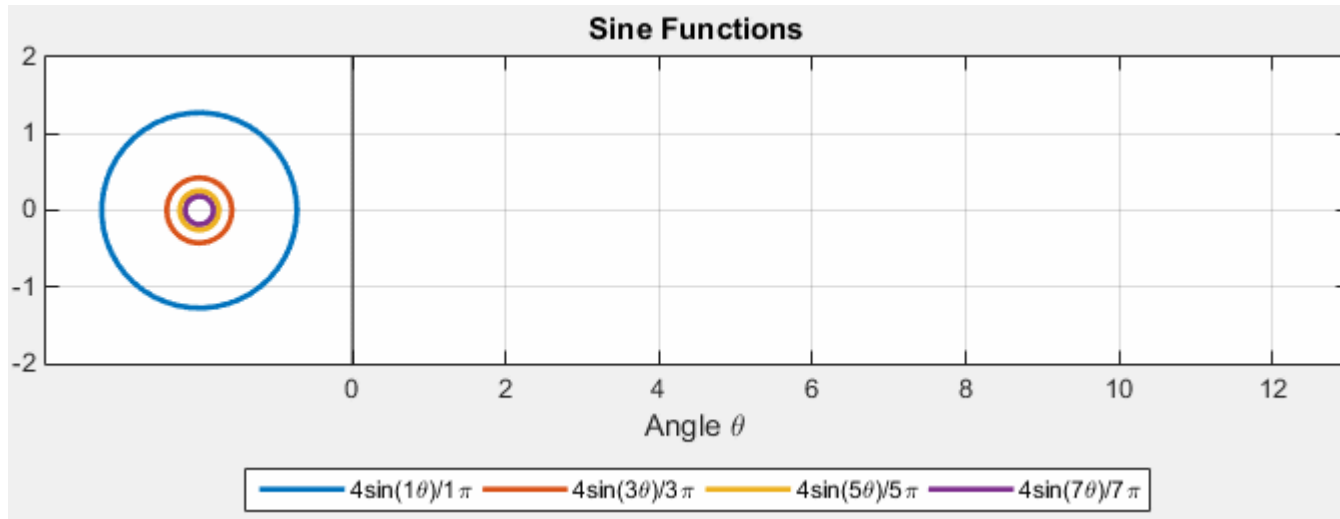
- Fourier series has many applications, infact any field of physical science that uses sinusoidal signals such as engineering, physics and applied mathematics.



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Applications of Fourier Series – Examples

- It can be used to approximate any signal. When we add sines and /or cosines the graph of cosines and sines become closer and closer to the original graph we are trying to approximate.



Applications of Fourier Series - Examples

- **Sound data compression often uses Fourier expansions.**
Take a sound, expand its Fourier series. The compression works by omitting higher terms of that expansion, which will make the signal "smoother", and containing less information to store.

Definition: Periodic function

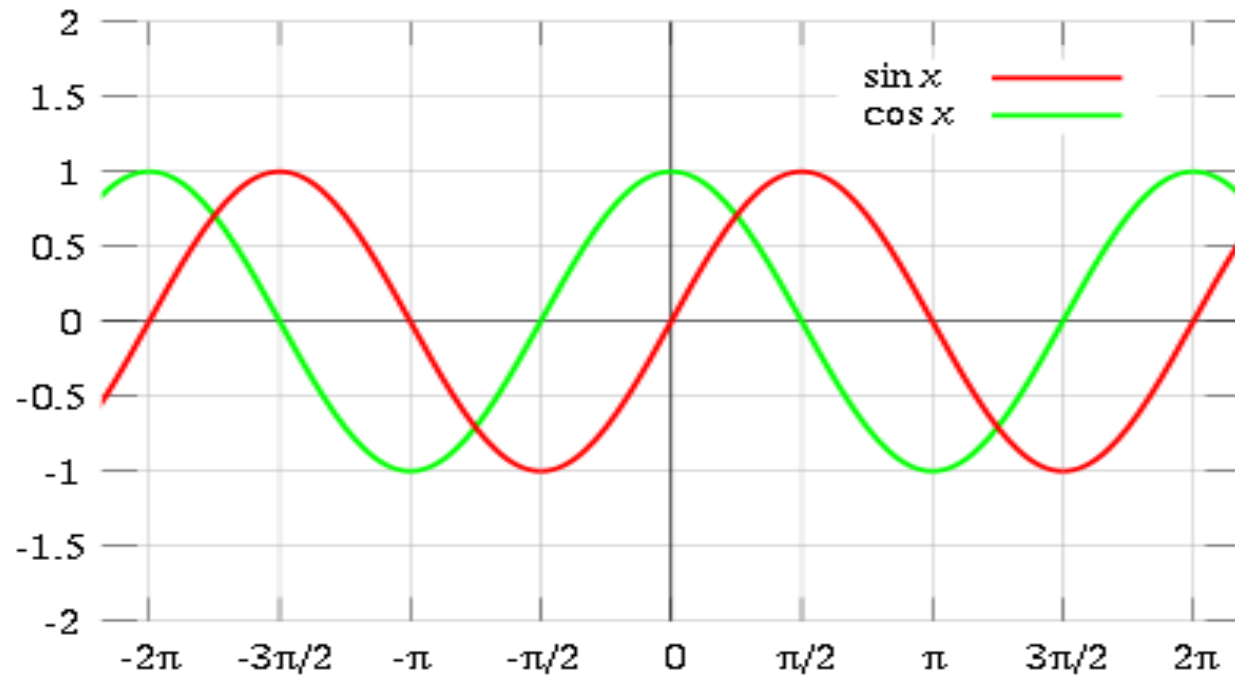
A function $f(x)$ is said to be periodic if $f(x + T) = f(x)$ for all real x and for some positive number T . Here T is known as the period of the $f(x)$.

❖ Fundamental period of $f(x)$ is the smallest period of f .

Eg. **$\cos x, \sin x$** ...are periodic functions with period 2π .

Result : If T is the period of $f(x)$, then nT is also period of f for any integer n .

Periodic function



Trigonometric Series

Trigonometric series is a fundamental series of the form

$$f(x) = a_0/2 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where constants a_0 , a_n and b_n are called the Euler coefficients.

Euler's (Fourier-Euler) Formulae

Let $f(x)$, a periodic function with period 2π defined in the interval $(c, c + 2\pi)$, be the sum of a trigonometric series,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (1)$$

$$\text{Then } a_0 = \frac{1}{\pi} \int_c^{c+2\pi} f(x) dx. \quad (2)$$

$$a_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos nx dx \text{ for } n = 1, 2, 3 \dots \quad (3)$$

$$b_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin nx dx \text{ for } n = 1, 2, 3 \dots \quad (4)$$

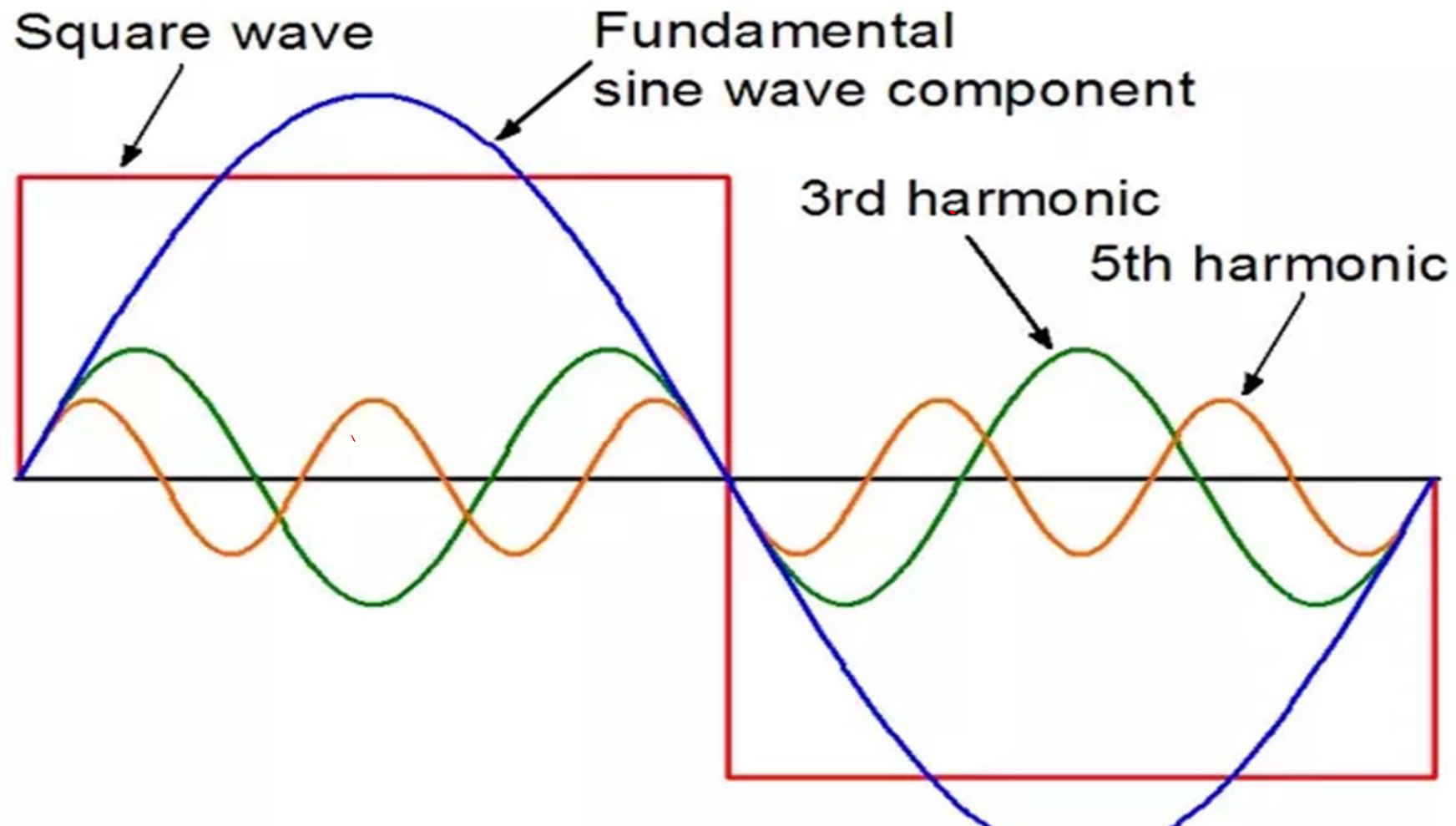
The formula (2),(3) and (4) are known as **Euler formulae** gives the coefficient's a_0, a_n and b_n which are known as **Fourier coefficients of $f(x)$** .

Definition: Fourier Series

Fourier series of a periodic function $f(x)$ with period 2π is the trigonometric series (1) with the Fourier coefficients a_0, a_n and b_n given by the Euler formulae (2),(3) and (4).

The individual terms in Fourier series is known as **harmonics**. The computation and study of Fourier series is known as harmonic analysis.

Harmonics



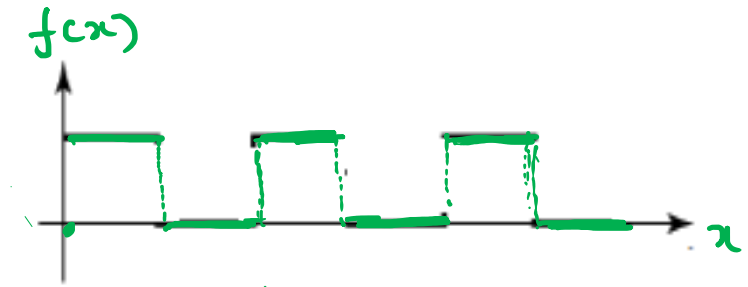
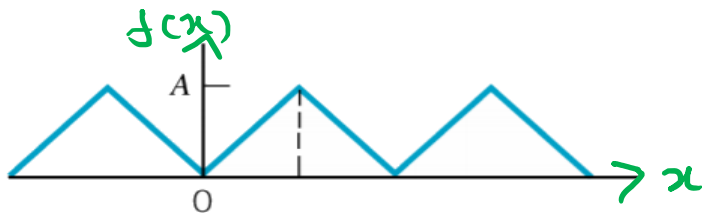
Existence of Fourier Series

Fourier has shown that the expansion of $f(x)$ as an infinite sum of sine and cosine waveforms is possible only if it satisfies certain conditions. These conditions are called Dirichlet conditions which are stated below:

Dirichlet Conditions:

Any function $f(x)$ defined in the interval $c < x < c + 2\pi$ with period 2π is said to satisfy the Dirichlet's conditions if,

- i) $f(x)$ is periodic and single valued function
- ii) $f(x)$ is continuous or $f(x)$ has only a finite number of discontinuities in a period of 2π
- iii) $f(x)$ has a finite number of maxima and minima in a given period.



Note: 1

To find Fourier coefficients a_0 , a_n , b_n when the interval is $(0, 2\pi)$.

If $c = 0$, then the interval of definition for $f(x)$ is $(0, 2\pi)$ and in this case

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \text{ for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \text{ for } n = 1, 2, 3 \dots$$

Note: 2

To find Fourier coefficients a_0 , a_n , b_n when the interval is $(-\pi, \pi)$

If $c = -\pi$, then the interval of definition for $f(x)$ is $(-\pi, \pi)$ and in this case

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \text{ for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \text{ for } n = 1, 2, 3 \dots$$

Note: 3

At the **point of discontinuity** x_0 , the Fourier series of $f(x)$ converges to the **arithmetic mean** of **left and right hand limit** of $f(x)$ at x_0 .

That is,

$$f(x_0) = \frac{1}{2} [f(x_0 - 0) + f(x_0 + 0)]$$

Some useful results in the computation of Fourier Series

Results:

1. $\cos n\pi = (-1)^n$

2. $\sin n\pi = 0$

3. $\cos \pi = -1$

4. $e^{-\infty} = 0$

5. $\sin \frac{n\pi}{2} = \begin{cases} 0, & n \text{ is even} \\ 1, & n = 1, 5, 9, \dots \\ -1, & n = 3, 7, 11, \dots \end{cases}$

6. $\cos \frac{n\pi}{2} = \begin{cases} 0, & n \text{ is odd} \\ 1, & n = 0, 4, 8, \dots \\ -1, & n = 2, 6, 10, \dots \end{cases}$

Results:

$$7. \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$8. \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

9. Bernoulli's Generalised Formula of Integration by Parts

If u and v are functions of x , then the Bernoulli's rule is

$$\int u * v dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

where $u', u'', u''', \dots$ are successive derivatives of u and $v_1, v_2, v_3, \dots$ are successive integrals of v



THANK YOU

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